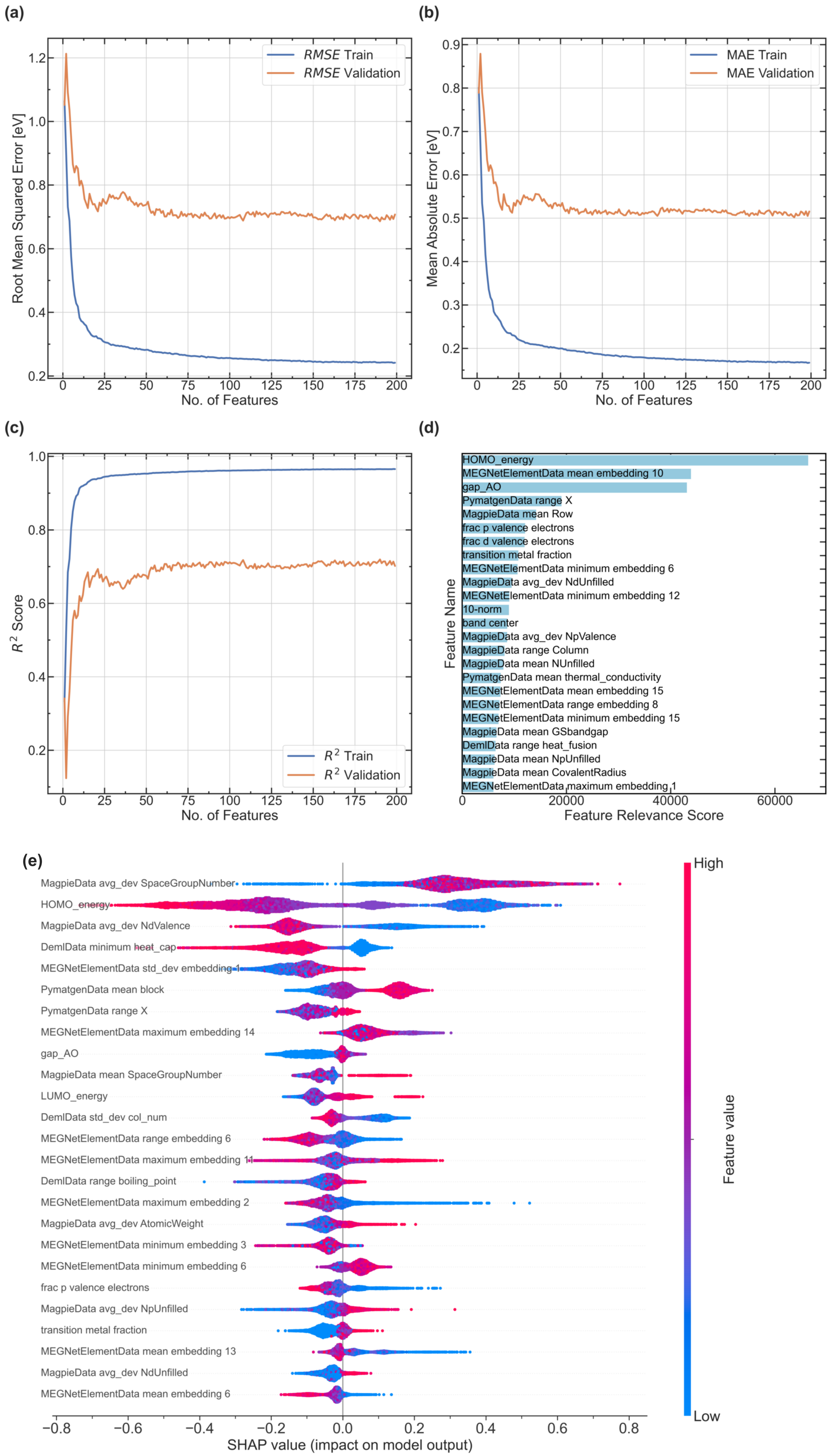
(a)
RMSE Train
RMSE Validation
Root Mean Squared Error [eV]
No. of Features
(b)
MAE Train
MAE Validation
Mean Absolute Error [eV]
No. of Features
(c)
R2 Train
R2 Validation
R2 Score
No. of Features
(d)
HOMO_energy
MEGNetElementData mean embedding 10
gap_AO
PymatgenData range X
MagpieData mean Row
frac p valence electrons
frac d valence electrons
transition metal fraction
MEGNetElementData minimum embedding 6
MagpieData avg_dev NdUnfilled
MEGNetElementData minimum embedding 12
10-norm
band center
MagpieData avg_dev NpValence
MagpieData range Column
MagpieData mean NUnfilled
PymatgenData mean thermal_conductivity
MEGNetElementData mean embedding 15
MEGNetElementData range embedding 8
MEGNetElementData minimum embedding 15
MagpieData mean GSbandgap
DemlData range heat_fusion
MagpieData mean NpUnfilled
MagpieData mean CovalentRadius
MEGNetElementData maximum embedding 1
Feature Name
Feature Relevance Score
(e)
MagpieData avg_dev SpaceGroupNumber
HOMO_energy
MagpieData avg_dev NdValence
DemlData minimum heat_cap
MEGNetElementData std_dev embedding 1
PymatgenData mean block
PymatgenData range X
MEGNetElementData maximum embedding 14
gap_AO
MagpieData mean SpaceGroupNumber
LUMO_energy
DemlData std_dev col_num
MEGNetElementData range embedding 6
MEGNetElementData maximum embedding 11
DemlData range boiling_point
MEGNetElementData maximum embedding 2
MagpieData avg_dev AtomicWeight
MEGNetElementData minimum embedding 3
MEGNetElementData minimum embedding 6
frac p valence electrons
MagpieData avg_dev NpUnfilled
transition metal fraction
MEGNetElementData mean embedding 13
MagpieData avg_dev NdUnfilled
MEGNetElementData mean embedding 6
SHAP value (impact on model output)
High
Feature value
Low